MATH 4780 / MSSC 5780 **Sec 102** - Regression Analysis

Fall 2026

Instructor: Dr. Cheng-Han Yu

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Office Hours: TuTh 3:20–4:50 PM or by appointment, Cudahy Hall 353
Help Desk/TA Hours: Not Available.

Website: math4780-f26.github.io/website/
Class Hours: TuTh 2–3:15 PM
Class Room: Cudahy Hall 118

1 Course Objectives

Starting from simple linear regression, this course provides a brief but complete foundation for understanding basic regression theory and applications. Regression is a core component of supervised learning in statistical machine learning. Topics include simple linear regression, multiple linear regression, diagnostic analysis, multicollinearity, nonparametric regression, variable selection, generalized linear models, and other selected topics.

2 Prerequisites

- MATH 2780, MATH 4720, or an equivalent introductory statistics course.
- Programming experience is helpful because the course involves doing regression analysis using R/Python programming language.
- Some exposure to linear algebra at the level of MATH 3100 is helpful because some matrix operations are used for modeling and computation.
- Prior coursework in MATH 4700 and MATH 4710 would also be helpful, but it is not required.
- Talk to me if you are not sure whether this is the right course for you.

3 Textbooks and References

- **[IS] (optional)** *Introduction to Statistics*, by Dr. Cheng-Han Yu. This is a useful resource for reviewing basic probability, statistics, and simple linear regression.
- **[LRA] (optional)** *Introduction to Linear Regression Analysis, 6th edition*, by D. C. Montgomery, E. A. Peck, and G. G. Vining. Publisher: Wiley. This is a graduate-level book that requires some knowledge of matrix algebra.
- **[CAR] (optional)** *An R Companion to Applied Regression*, by John Fox and Sanford Weisberg. Publisher: SAGE. This book makes regression analysis in R easier, especially through the `car` and `effects` packages.

- **[RD] (optional)** *Regression Diagnostics*, by John Fox. Publisher: SAGE. This is a concise book for learning regression diagnostics.
- **[CMR] (optional)** *Classical and Modern Regression with Applications*, by Raymond H. Myers. Publisher: Duxbury Press. This was one of the textbooks I used when I was a master's student.
- **[AR] (optional)** *All of Regression*, by Isabella Verdinelli and Larry Wasserman. Publisher: Cambridge University Press. This is a clear and concise graduate-level book that covers many modern regression topics.

4 Course Management

- All course materials are posted on our course website https://math4780-f26.github.io/website/.
- Course grades are saved and managed in **D2L** > **Assessments** > **Grades**.
- Coursework is submitted to **D2L** > **Assessments** > **Dropbox**.
- Check **D2L** > **News** for new announcements.

5 Programming Languages

- You may use any programming language to do you work.

6 Email Policy

- I will attempt to reply to your email quickly, usually **within 24 hours**.
- **Expect a reply on Monday if you send a question over the weekend**. If you do not receive a response from me within two days, resend your question or comment in case there was a "mix-up" with email communication. I hope this will not happen.
- Please start your email subject line with **[math4780]** or **[mssc5780]** followed by a clear description of your question. See an example in Figure 1.

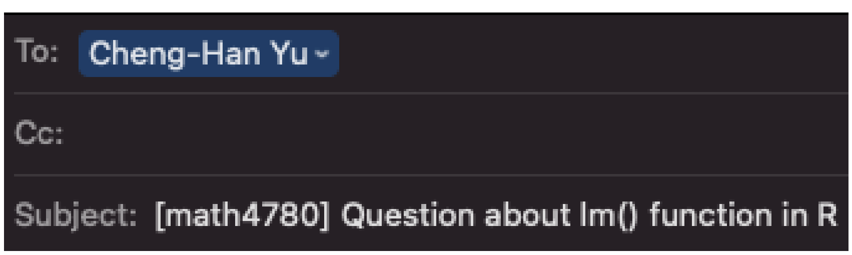

Figure 1: Email Subject Line Example

- Email etiquette is important. Please read this article to learn more about email etiquette.
- I am more than happy to answer your questions about this course or statistics in general. However, due to time constraints, I may choose **NOT** to respond to students' emails if
  1. The student could answer their own inquiry by reading the syllabus or information on the course website or D2L.

2. The student is asking for an extra-credit opportunity. The answer is “no”.
3. The student is requesting an extension on homework. The answer is “no”.
4. The student is asking for a grade to be raised for no legitimate reason. The answer is “no”.
5. The student is sending an email without proper etiquette.

7 Grading Policy

Students in this class may range from undergraduate students to graduate students, so it makes sense to evaluate student performance using different point scales for MATH 4780 and MSSC 5780.

For students in **MATH 4780**, the final grade is earned out of **1000 total points** distributed as follows:

Category	Points
Homework 1 to 6	150
In-class exam	250
Final project	400
Class activity	200
Total	**1000**

For students in **MSSC 5780**, the final grade is earned out of **1200 total points** distributed as follows:

Category	Points
Homework 1 to 6	150
In-class exam	250
Final project	400
Class activity	200
MSSC work	200
Total	**1200**

- You will **NOT** be allowed to complete any extra-credit projects, homework, or exams to compensate for a poor average. Everyone must be given the same opportunity to do well in this class. Individual exams will **NOT** be curved.
- The final grade is based on your percentage of points earned out of 1000 or 1200 points and the grade-percentage conversion table. $[x, y)$ means greater than or equal to x and less than y. For example, 94.1 is in $[94, 100]$, so the grade is A, and 92.8 is in $[90, 94)$, so the grade is A-.

Table 3: Grade-Percentage Conversion

Grade	Percentage
A	[94, 100]
A-	[90, 94)
B+	[87, 90)
B	[83, 87)
B-	[80, 83)

Grade	Percentage
C+	[77, 80)
C	[73, 77)
C-	[70, 73)
D+	[65, 70)
D	[60, 65)
F	[0, 60)

- This is not a course in which most students receive a grade of A. If you want to earn a good grade, study hard. No pain, no gain.

8 Homework

- Homework will be assigned through the course website.
- To submit your homework, please go to **D2L** > **Assessments** > **Dropbox** and upload your homework in **PDF** format.
- There will be 6 graded homework sets: Homework 1 to Homework 6.
- Some questions are required for MSSC 5780 students.
- **No late homework will be accepted, and you will not be allowed to make up missed homework.**
- **Handwriting is not allowed for the data analysis parts.**
- Generative AI (GenAI) is allowed for homework, but you **must carefully cite it or reveal your use of AI.** See Section 13 for more details.

9 In-Class Exam

- The in-class exam covers materials from **Weeks 1 to 10** on simple and multiple linear regression.
- Some questions are required for MSSC 5780 students.
- **No make-up exam** will be given unless you have an excused absence.

10 Class activity

- You will be presenting exercise problems or mini projects.
- More details will be released later.

11 Final Project

- A detailed final project guideline will be released later.
- Generative AI (GenAI) is allowed for the final project, but you **must carefully cite it or reveal your use of AI.** See Section 13 for more details.

12 MSSC Work

- MSSC 5780 students will complete additional work worth 200 points.
- Details will be released later.

13 Generative Artificial Intelligence (GenAI) Policy

- *You are responsible for the content of all work submitted for this course.*
- For your homework, final project, and other take-home work, you are allowed to use generative AI tools such as ChatGPT to generate a draft of your work.
- To avoid academic integrity issues, you **must cite your use of AI or screenshot your entire AI usage history**. Check the following resources for guidance on how to cite it.
  - How to cite ChatGPT
  - How to Cite AI-Generated Content
- If you use GenAI, please include the following in your submitted work:
  - **How I used AI (prompts or questions)**
  - **Generated output (screenshot or copy-paste excerpt)**
  - **How I used the output**

Here is an example.

- **How I used AI (prompts and questions)**
  - *I asked ChatGPT to generate a histogram using R.*
- **Generated output (screenshot or copy-paste excerpt)**

GenAI screenshot example goes here.

- **How I used the output**
  - *I reviewed the suggestions, but I did not use the exact code. Instead, I changed the code format and set the breaks value to 50.*

14 Academic Integrity

- Watch this video about academic integrity issues related to using GenAI.
- This course expects all students to follow University and College statements on academic integrity.
- **Honor Pledge and Honor Code**: *I recognize the importance of personal integrity in all aspects of life and work. I commit myself to truthfulness, honor, and responsibility, by which I earn the respect of others. I support the development of good character, and commit myself to uphold the highest standards of academic integrity as an important aspect of personal integrity. My commitment obliges me to conduct myself according to the Marquette University Honor Code.*

15 Accommodation

If you need to request accommodations, or modify existing accommodations that address disability-related needs, please contact Disability Services.

16 Tentative Course Schedule

Week 1, 8/31 - 9/6: Syllabus; Overview of Regression; Probability and Statistics Review

Week 2, 9/7 - 9/13: Simple Linear Regression

- **Labor Day (9/7)**
- **Drop deadline by 11:59 PM on Tuesday, 9/8**
- **Homework 1 due by 11:59 PM on Friday, 9/11**

Week 3, 9/14 - 9/20: Simple Linear Regression

Week 4, 9/21 - 9/27: Multiple Linear Regression

- **Homework 2 due by 11:59 PM on Friday, 9/25**

Week 5, 9/28 - 10/4: Multiple Linear Regression

Week 6, 10/5 - 10/11: Multiple Linear Regression in Matrix Form (MSSC)

- **Homework 3 due by 11:59 PM on Friday, 10/9**

Week 7, 10/12 - 10/18: Regression Diagnostics: Residuals and Unusual Data

Week 8, 10/19 - 10/25: Regression Diagnostics: Symmetry, Normality, and Constant Variance

- **No class on Thursday, 10/22 (Midterm Break)**

Week 9, 10/26 - 11/1: Regression Diagnostics: Linearity and Lack of Fit

- **Midterm grade submission by noon on Tuesday, 10/27**
- **Homework 4 due by 11:59 PM on Friday, 10/30**

Week 10, 11/2 - 11/8: Bootstrapping and Simulation-Based Inference

Week 11, 11/9 - 11/15: Polynomial and Nonparametric Regression

- **In-class Exam on Tuesday, 11/10**

Week 12, 11/16 - 11/22: Categorical Variables

- **Withdrawal deadline on Friday, 11/20**
- **Homework 5 due by 11:59 PM on Friday, 11/20**

Week 13, 11/23 - 11/29: Collinearity

- **No class on Thursday, 11/26 (Thanksgiving)**

Week 14, 11/30 - 12/6: Variable Selection

Week 15, 12/7 - 12/13: Logistic Regression

- **Homework 6 due by 11:59 PM on Friday, 12/11**

Week 16, 12/14 - 12/20: Final Project

- **Final Project Presentation on MONDAY, 12/14, 10:30 AM - 12:30 PM**

Week 17, 12/21 - 12/27

- **Final grade submission by noon on Tuesday, 12/22**

Dr. Yu reserves the right to change the syllabus.